RISE and Yocto: Building a RISC-V Board Farm

What Is The Yocto Project?

1. **Versatile: An ecosystem** of flexible tools and configurations **for building custom Linux distributions**
2. **Open Source:** A **collaborative project** between leading companies and organizations - **including RISC-V International**
3. **Compatible: Supports RISC-V as a first-class architecture** on hardware (e.g. SpacemiT K1) via the `meta-riscv` layer (**SpacemiT K3 support in progress**) - and QEMU (`qemuriscv64`, **RVA23S64** profile)

The Yocto Project

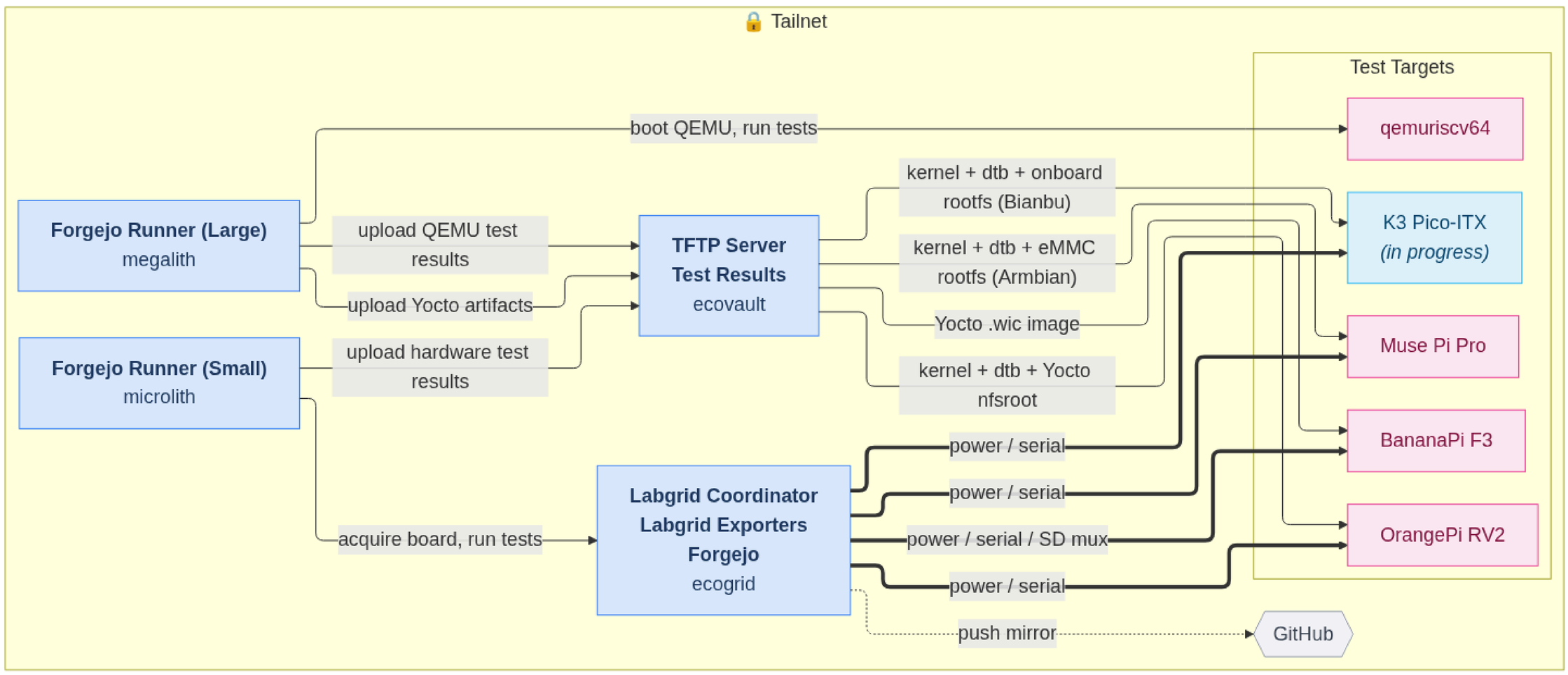
meta-riscv layer

Why Build A Board Farm?

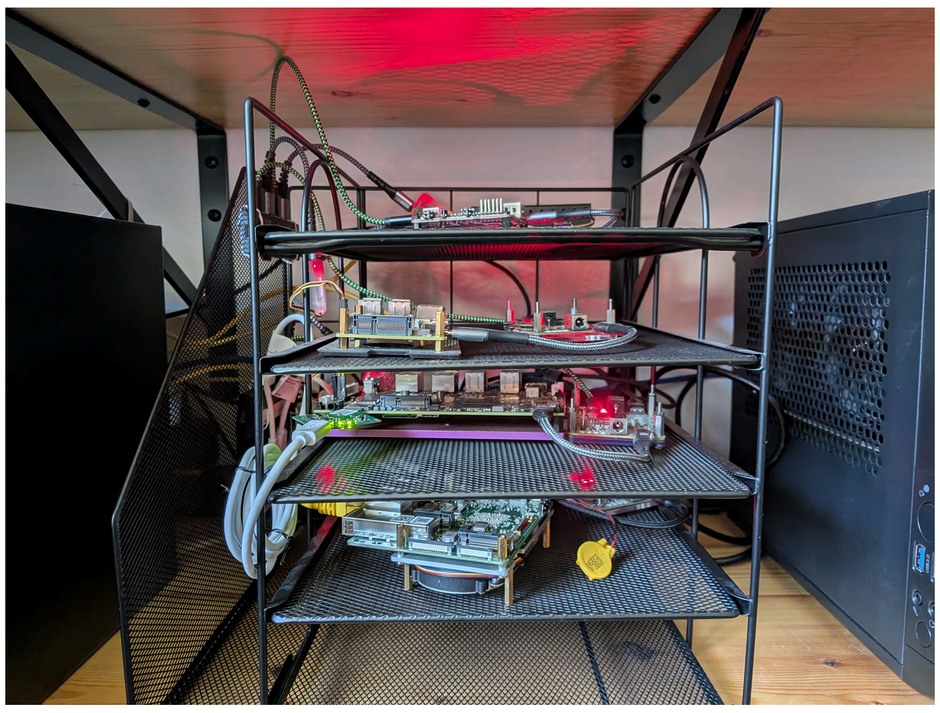
1. **Simplify Maintenance:** Nightly testing **catches regressions early** for RISC-V BSPs and critical packages in the Yocto ecosystem
2. **Extensible Configuration:** Automation logic for Yocto **can easily be repurposed** for the Linux kernel and RISE's broader efforts
3. **Create a Reference Design:** `boardgarden` provides a **blueprint** to combine **open-source CI/CD (Forgejo)** and **remote hardware control (Labgrid)** for your automated testing needs. **See more here:**

boardgarden repository

`boardgarden`: Network Diagram



A Look At The Hardware



Top to Bottom: OrangePi RV2, Muse Pi Pro, BananaPi F3, K3 Pico-ITX (with room for more!)

What's Next?

- **Add more target devices** - K3 platforms, SiFive P570D, others TBD
- **Broaden testing scope** in `boardgarden` - more hardware configurations (e.g. different storage options, attached devices).
- **Include more Yocto layers**, **upstream kernel builds**, add **benchmarking**, etc.
- **Create a `genericriscv64` BSP** for the Yocto Project core, **and incorporate that into the `boardgarden` test matrix**
- **Teach others** how to build robust, automated test setups using this work as a reference
- **Expand** outside of the Yocto kernel + bootloader + rootfs testing scope - **by targeting RISE's Python packaging achievements** (link below) and other areas of interest

`wheel_builder` (Python for `riscv64`)

Acknowledgements

Thanks to the RISE Project for providing BayLibre with the opportunity to improve the RISC-V support of the Yocto Project, and to the many upstream maintainers whose efforts have made building our automated test infrastructure possible. We look forward to the future of RISC-V in the Yocto Project!

Learn more about BayLibre!

Learn more about RISE!

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